1 Course links

The course syllabus lists the reading for every week, and questions go to the instructor by email. Last year's notes are at `https://example.org/physics/archive/2025/lecture-notes/week-01-kinematics.pdf` and the problem sets are here. The lab data is on an old file server that the PDF still links to.

2 Motion with constant acceleration

A cart that starts with speed v_0 and speeds up at a steady rate a has, after a time t, the speed

$$v = v_0 + a t \tag{1}$$

and has covered the distance

$$x = v_0 t + \tfrac{1}{2} a t^2 . \tag{2}$$

Eliminating t between Eqs. (1) and (2) gives the third relation of the section. Equation (1) holds for any steady rate, and so does Eq. 2; Figure 1 shows the cart, and Figure 1 links to it from the PDF itself.

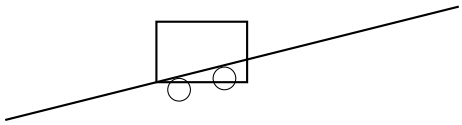

Figure 1: A cart on a ramp.

3 A note in French

The exam has one question written in French, quoted here as it will be printed:

> La vitesse moyenne est le quotient du déplacement par la durée du trajet, et elle s'exprime en mètres par seconde.

4 Colour in the notes

In these notes, the vectors are printed in red and scalars in black, so read the terms shown in red as vectors. A light grey hint follows each problem. This sentence is dark grey, which reads as black.